Yes, the **Lattice Radiant Software** includes a feature called **Reveal**, which is a logic analysis tool that allows you to probe and debug internal and external signals, including those on the physical pins of a Lattice FPGA, such as the **CrossLink-NX**. The Reveal tool is designed for real-time signal monitoring and debugging, enabling you to observe the behavior of signals directly on the FPGA's I/O pins or internal logic. Below is a detailed explanation of how Reveal can probe real signals on pins and the steps to set it up, particularly in the context of a design involving a HyperRAM like the **Winbond W958D6NBKX**.

Can Reveal Probe Real Signals on Pins?

- **Yes**, Reveal can probe signals on the FPGA's physical I/O pins (external signals) as well as internal signals within the FPGA's logic. This includes signals like those used in a HyperBus interface (e.g., CS#, CK, CK#, DQ[7:0], RWDS for the W958D6NBKX).
- **How it Works**: Reveal uses an embedded logic analyzer (Reveal Analyzer) inserted into the FPGA design to capture and display signal activity in real time. You can configure it to monitor specific I/O pins or internal nodes, and the captured data is sent to the Radiant software via a JTAG interface for analysis.
- **Limitations**:
  - Probing is limited by the FPGA's resources (e.g., block RAM for trace storage) and the number of signals you can monitor simultaneously.
  - External pins must be accessible and configured in the design (e.g., assigned to specific I/O standards like LVCMOS18 for HyperRAM).
  - High-speed signals (e.g., 200 MHz HyperBus) may require careful setup to ensure accurate capture, as the Reveal Analyzer's sampling rate is limited by the FPGA's internal clocking resources.

Steps to Probe Real Signals on Pins Using Reveal in Lattice Radiant Software:

1. **Set Up Your Design in Radiant**:
    - Create or open your FPGA design (e.g., a HyperBus controller for the W958D6NBKX) in Lattice Radiant.
    - Ensure the I/O pins are correctly assigned in the Pin Constraints Editor (.pcf file) for the signals you want to probe (e.g., CS#, CK, DQ[7:0], RWDS for HyperRAM). For the W958D6NBKX, these pins should be configured for LVCMOS18 (1.8V single-ended signaling, as discussed in prior responses).
    - Synthesize and map the design to the target FPGA (e.g., CrossLink-NX).
2. **Insert the Reveal Core**:
    - In Radiant, open the **Reveal Inserter** tool from the Tools menu.
    - Create a new Reveal project or add Reveal to your existing design.
    - Select the signals to probe:
        - **External I/O Pins**: Choose the physical pins (e.g., those assigned to HyperRAM signals like DQ[7:0] or RWDS). These are listed based on your .pcf file.
        - **Internal Signals**: Optionally, select internal nets (e.g., signals within your HyperBus controller) for deeper debugging.
    - Configure the Reveal core:
        - Set the **sample depth** (number of samples to capture, limited by FPGA block RAM).
        - Set the **trigger conditions** (e.g., trigger on CS# going low to capture a HyperRAM read of the ID0 register).
        - Specify the **sample clock** (e.g., use the HyperBus clock CK or a slower internal clock, ensuring it’s fast enough to capture 200 MHz signals accurately).
    - Insert the Reveal core into the design. This modifies the FPGA design to include the logic analyzer.

3. **Synthesize and Implement the Design**:
   - After inserting the Reveal core, re-synthesize, place, and route the design in Radiant.
   - Ensure the design meets timing requirements, especially for high-speed HyperBus signals (e.g., 200 MHz for the W958D6NBKX). Use Radiant's Timing Analyzer to verify.
4. **Program the FPGA**:
   - Program the FPGA (e.g., CrossLink-NX) with the modified bitstream containing the Reveal core using a JTAG interface and a supported programmer (e.g., Lattice Diamond Programmer or Radiant's integrated programmer).
5. **Run the Reveal Analyzer**:
   - Open the **Reveal Analyzer** tool in Radiant.
   - Connect to the FPGA via JTAG.
   - Configure the trigger conditions (e.g., trigger on a specific HyperRAM command, such as the CS# low for an ID0 register read).
   - Start the capture process. The Reveal Analyzer will monitor the selected pins (e.g., DQ[7:0] for ID0 data output) in real time and store the data in the FPGA's block RAM.
   - Once triggered, the captured signal waveforms are uploaded to the Radiant software for display.

6. **Analyze the Signals**:
    - View the captured waveforms in the Reveal Analyzer's waveform viewer.
    - For the W958D6NBKX ID0 register read:
        - Check the CS# signal to confirm the transaction start.
        - Verify the 48-bit Command-Address (CA) packet on DQ[7:0] (e.g., `0xE000000000` for register read at address 0x0000).
        - Observe RWDS to identify valid data output.
        - Confirm the ID0 data (e.g., Winbond's Manufacturer ID `0x0B` and Device ID) on DQ[7:0] after the latency period.
    - Use the waveform to debug timing issues, signal integrity, or protocol errors (e.g., incorrect latency or data misalignment).

Example for W958D6NBKX HyperRAM:

To probe the ID0 register read (as described in the prior response):

- **Signals to Probe**:
    - CS# (chip select, active low).
    - CK, CK# (differential clock).
    - DQ[7:0] (data lines for CA packet and ID0 data).
    - RWDS (read/write data strobe, high during valid data output).
- **Trigger Setup**: Trigger on CS# going low to capture the HyperBus transaction.
- **Expected Output**: After the CA packet (`0xE000000000`) and latency (e.g., 6 cycles at 200 MHz), the ID0 register should output 2 bytes (e.g., `0x0BXX`, where `0x0B` is Winbond's JEDEC ID and `XX` is the device-specific ID, per the W958D6NBKX datasheet).
- **Clock**: Use the 200 MHz HyperBus clock (CK) or a divided clock for sampling, ensuring it meets the Reveal Analyzer's timing constraints.

Key Considerations:

- **Resource Usage**: The Reveal core consumes FPGA resources (e.g., LUTs, block RAM). Ensure your CrossLink-NX design has enough resources available.
- **Sampling Rate**: For a 200 MHz HyperRAM interface, the Reveal Analyzer's sampling clock must be fast enough to capture signal transitions accurately. If the internal clock is slower, you may miss high-speed transitions.
- **Pin Access**: Ensure the I/O pins are not multiplexed or obstructed by other functions in the design. Verify pin assignments in the .pcf file.
- **JTAG Connection**: A stable JTAG connection (e.g., via a Lattice HW-USBN-2B cable) is required for real-time data transfer to the Reveal Analyzer.
- **Documentation**: Refer to the **Lattice Radiant Reveal User Guide** (available on www.latticesemi.com) for detailed setup instructions and limitations.

Resources:

- **Lattice Radiant Software**: Download the latest version from www.latticesemi.com and access the Reveal Inserter and Analyzer tools.
- **Reveal User Guide**: Available in the Radiant software documentation or on Lattice's website, detailing configuration steps.
- **Community Support**: Check community.latticesemi.com for examples of Reveal usage with CrossLink-NX or similar FPGAs.
- **W958D6NBKX Datasheet**: Confirm the ID0 register format and HyperBus timing requirements from Winbond's website (www.winbond.com).

Conclusion:

The Lattice Radiant Software's **Reveal** tool can effectively probe real signals on the FPGA's I/O pins, such as those used for the W958D6NBKX HyperRAM interface (e.g., CS#, DQ[7:0], RWDS). By inserting a Reveal core and configuring the Analyzer, you can monitor HyperBus transactions, including the ID0 register read, in real time. Ensure proper pin configuration (LVCMOS18 for 1.8V), adequate sampling clock, and sufficient FPGA resources for accurate debugging.

If you need specific guidance on setting up Reveal for your CrossLink-NX design or analyzing HyperRAM signals, please provide additional details (e.g., your design's HDL or specific signals to probe), and I can offer tailored instructions.